The Effect of Emigration on Unemployment in the Grand Duchy of Luxembourg

Early Results

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Rapid review the research question & project positioning

- Research question: "**How does emigration affect unemployment in Luxembourg?**"
- Luxembourg has a highly **international workforce** (almost **50%** of workers are non-Luxembourgers).
- Two possible effects of emigration:
  1. **Positive impact on unemployment**: Fewer workers → more vacancies filled by unemployed locals → lower unemployment.
  2. **Negative impact on unemployment**: Skilled labour shortages → employers struggle to fill roles.
- The findings could inform Luxembourg's **labour migration policies**, by seeing whether **emigration** creates **labour gaps** or relieves pressure on the job market.

Detailed presentation of the research methodology, including data & model

Empirical Approach

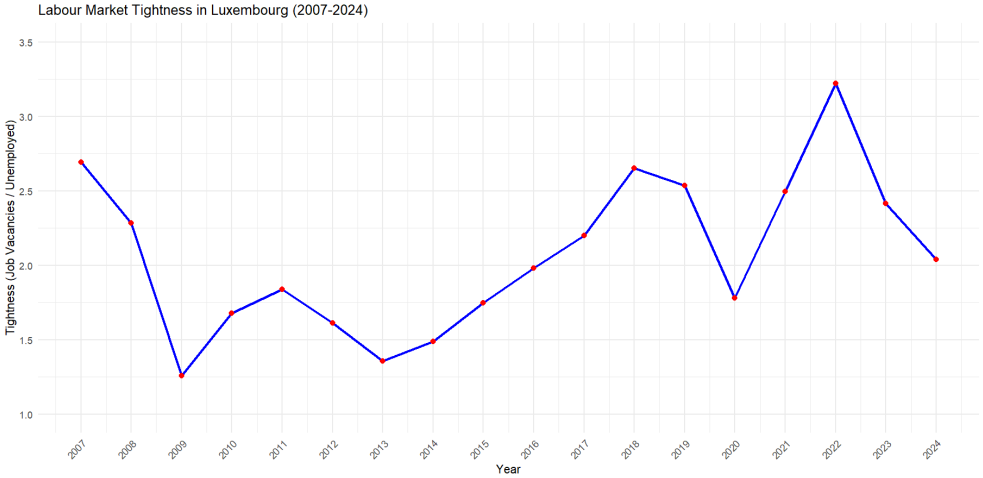
- **Labour Market Tightness** (Job Vacancies / Unemployment) Over Time → (key graph & equation determining job-finding rate, job-filling rate, welfare etc).
- **Migration Trends** → Arrivals vs. Departures vs. Net Migration Over Time.
- **Arrivals/Departures/Net Migration trend** for Luxembourg, Belgium, France, Germany, Italy, Spain, and Portugal.
- **Labour Market Tightness & Net Migration Over Time.**
- **Unemployment & Net Migration Over Time**.
- **Labour Market Tightness & Portugal's Net Migration Over Time.**
- **VAR Model (Vector AutoRegression)** → Tests how emigration and unemployment interact over time. Testing the **statistical link** between emigration and unemployment.
- **Granger Causality Test** → Checks if migration "causes" unemployment changes, or vice versa.

Detailed presentation of the research methodology, including data & model

Data Collection

- **Sources**: STATEC (Luxembourg's national statistics office) – Unemployment & vacancies.
- ADEM (Luxembourg's employment agency) – Job postings & workforce data.
- Eurostat & World Bank – Migration & labor force statistics.
- **Data:** Unemployment rates, vacancy rates, emigration trends.

Detailed presentation of the early results, with tables & graphs



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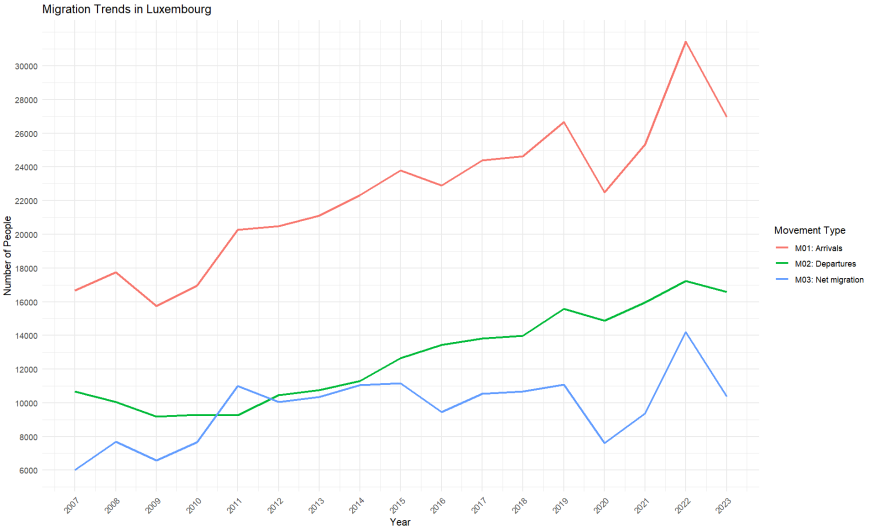

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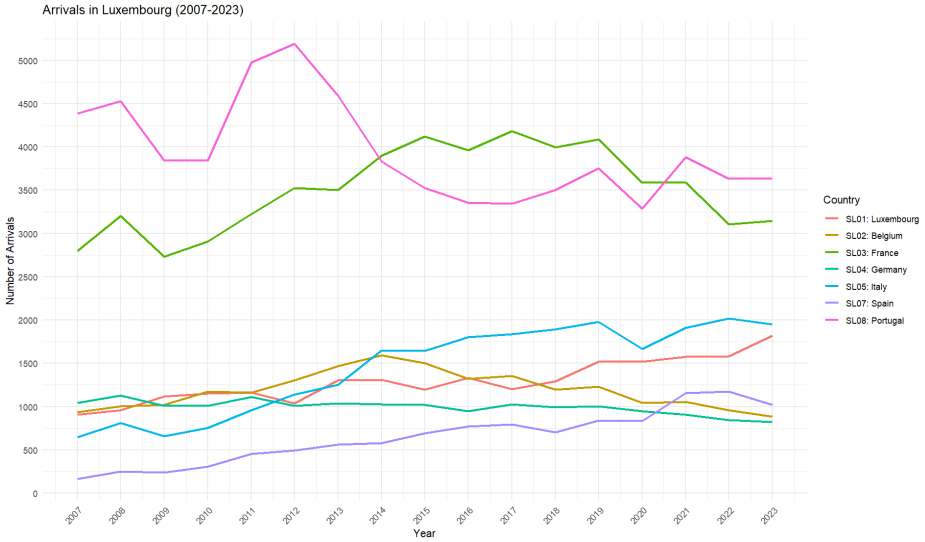

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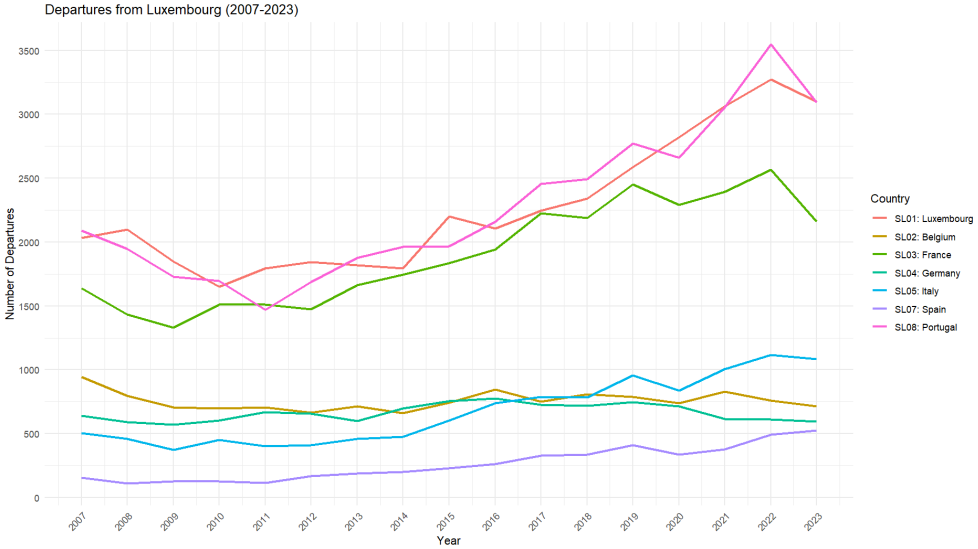

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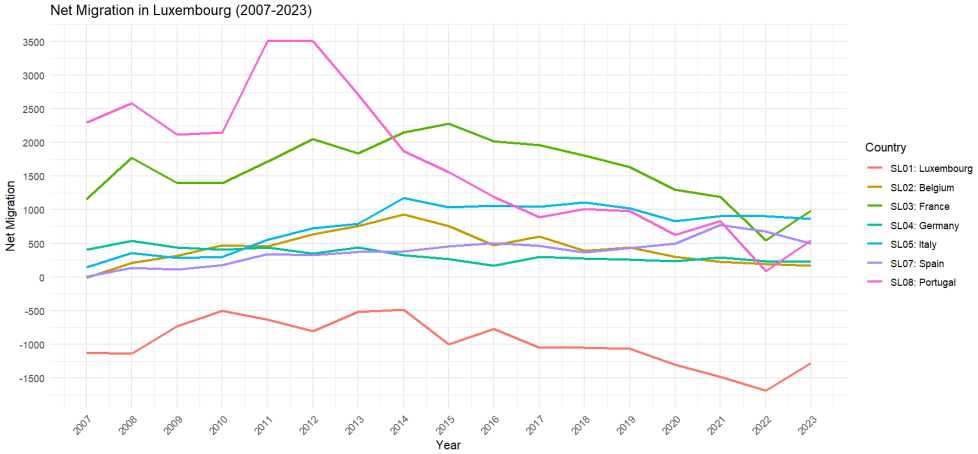

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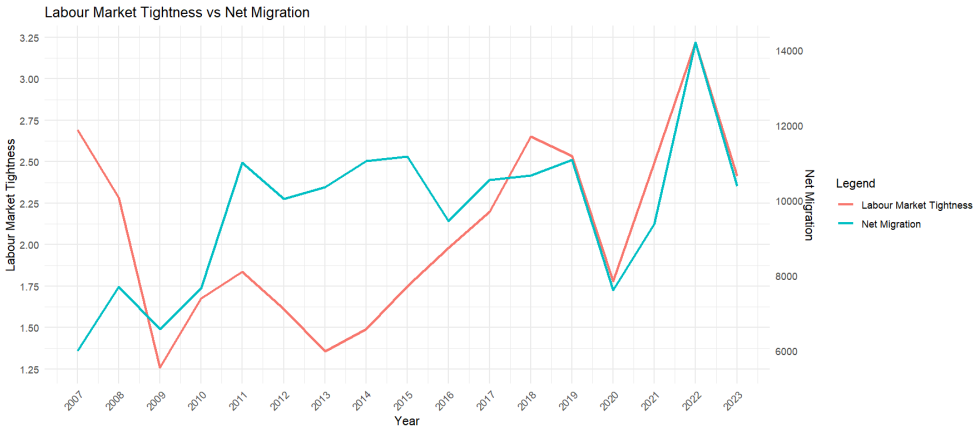

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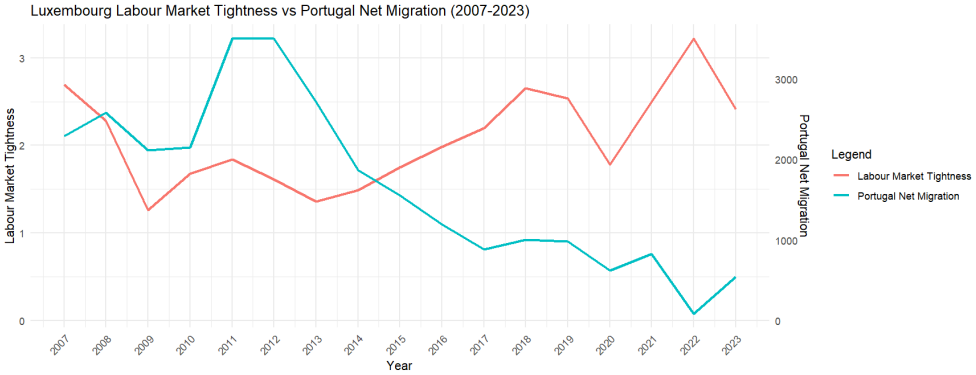

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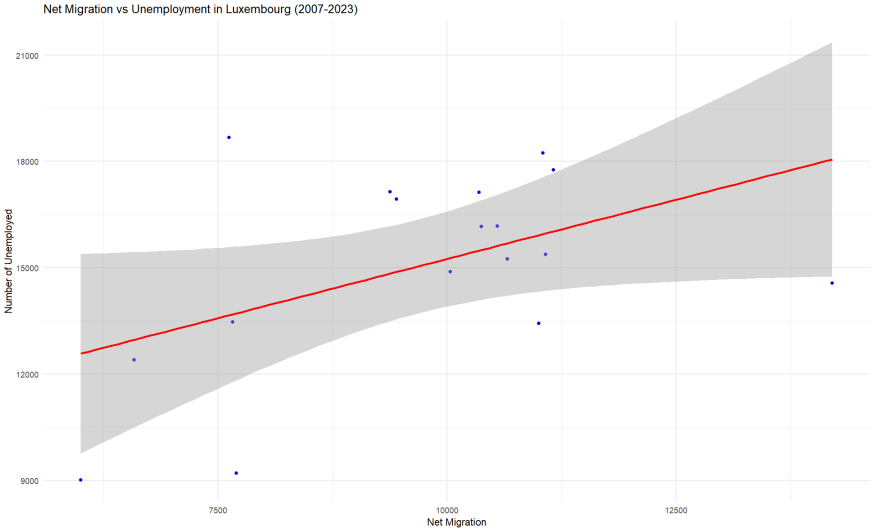

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Granger Causality Test Results

Direction	F_Statistic	P_Value	Significance
Net Migration → Unemployment	4.948719	0.0444432	* Significant
Unemployment → Net Migration	5.292105	0.0386260	* Significant

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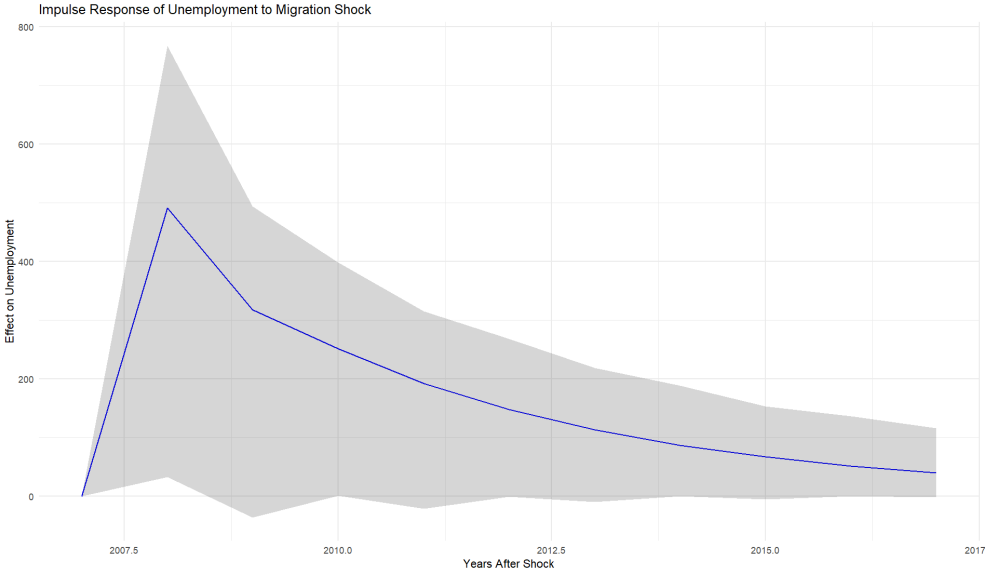

Existing research hurdles & next steps

Hurdles:

- **Data availability**: Extremely difficult to find Labour Market Tightness data on the **other countries**, and to find **sector-specific data** in Luxembourg.
- **Causality issue**: Unemployment & migration **move together**, but why? Do workers leave because jobs are scarce, or does emigration create skill shortages that increase unemployment?
- Hard to distinguish **voluntary vs. forced emigration.**
- **Policy Distortions:** Government interventions (e.g. tax incentives) may alter migration trends, affecting findings.

Existing research hurdles & next steps

Next Steps:

1. **Explore Sectoral Trends:** Investigate whether **certain industries** are more impacted by migration shifts.
2. Plot a **Relative Labour Market Tightness graph** for Luxembourg and the other countries, and see if changes in Luxembourg's labour market tightness influence labour market conditions in surrounding countries, particularly those with a high percentage of cross-border workers (e.g., Portugal, France, Belgium, Germany).
3. Draft **final findings** for review and refinement.